ASSIGNMENT

Programming Fundamentals Lab-2

TASK 01:

1-
```
# include <iostream>
Using namespace std;
int main ( )
{
char letter;
letter = 'A';
Cout << letter << '\n';
letter = 'B';
cout << letter << '\n';
return 0;
}
```

Output:

A
B

2-
```
# include <iostream>
Using namespace std;
int main ( )
{
char letter;
letter = 65;
cout << letter << '\n';
letter = 66;
Cout << letter << '\n';
Cin >> letter;
return 0;
}
```

Output

A
B

3-
```
# include <iostream>
Using namespace std;
int main ( )
{
int letter;
letter = 'A';
Cout << letter << '\n';
letter = 'B';
Cout << letter << '\n';
Cin >> letter;
return 0;
}
```

Output:

65
66

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Coding C++

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RUN MENU

```
1
6       char letter;
7
8       // Part 1
9       letter = 'A';
10      cout << letter << endl;
11
12      letter = 'B';
13      cout << letter << endl;
14
15      // Part 2
16      letter = 65;
17      cout << letter << endl;
18
19      letter = 66;
20      cout << letter << endl;
21
22      // Part 3
23      int number;
24
25      number = 'A';
26      cout << number << endl;
27
28      number = 'B';
29      cout << number << endl;
30
31      return 0;
32  }
```

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Compile Result

```
A
B
A
B
65
66

[Process completed - press Enter]
```

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TASK 02:

```
# include <iostream>
Using namespace std;
int main ( )
{
int num1, num2;
Cout << Enter two integers:";
Cin >> num 1 >> num 2;
cout << "Addition=" << num1 + num2 << endl;
cout << "subtraction =" << num1 - num 2 << endl;
Cout << "Multiplication =" << num 1 * num 2 << endl;
cout << "Division=" << num 1/num2 << endl;
return 0;
}
```

Output:

Enter two integers : 20 5
Addition = 25
subtraction = 15
Multiplication = 100
Division = 4

TASK 03:

```
# include "stdafx.h"
# include <iostream>
Using namespace std;
int main ( )
[
int val 1;
int val 2;
int val 3;
Cout << Plese enter your 3 numbers:";
Cin >> val 1 >> val 2 >> Val 3;
Cout << val1 << "\n" << val 2 << "\n" << val 3 << "\n";
Cout << val3 << "\n" << val 2 << "\n" << val 1 << "\n";
system ("PAUSE");
return 0;
}
```

Output:

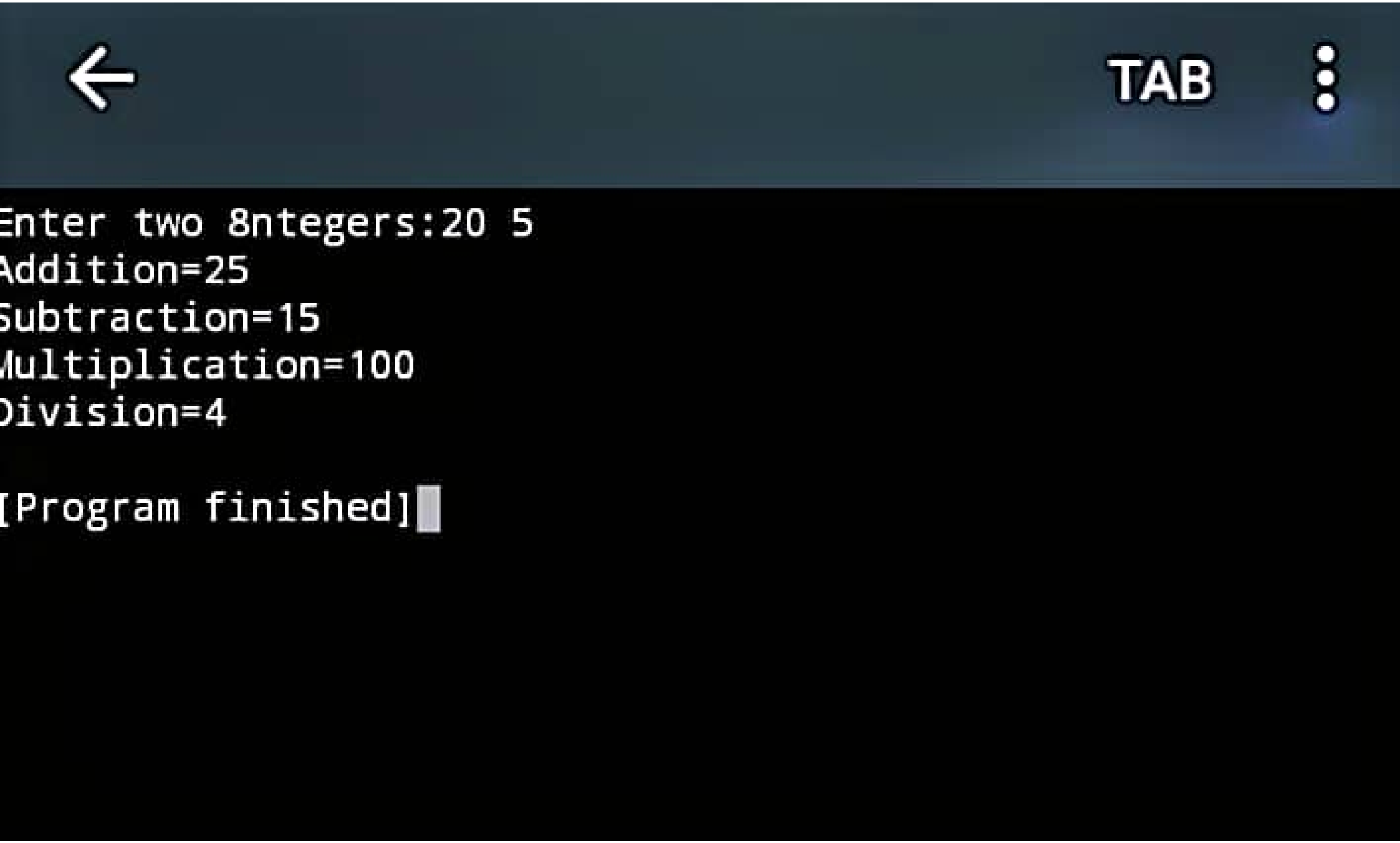
Plese enter your 3 numbers:
10
20
30
30
20
10

CamScanner

new*

```
1  # include <iostream>
2  using namespace std;
3  int main ( )
4  {
5  int num1,num2;
6  cout<<"Enter two 8ntegers:";
7  cin>>num1>>num2;
8  cout<<"Addition="<<num1+num2<<endl;
9  cout<<"Subtraction="<<num1-
   num2<<endl;
10 cout<<"Multiplication="<<num1*
   num2<<endl;
11 cout<<"Division="<<num1/num2<<endl;
12 return 0;
13 }
14
```

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TAB
Enter two 8ntegers:20 5
Addition=25
Subtraction=15
Multiplication=100
Division=4
[Program finished]

CamScanner


Coding C++
RUN
MENU
Auto saved at 01:52:37
#include <iostream>
using namespace std;
int main()
{
int val1;
int val2;
int val3;
cout << "Please enter your 3 numbers: ";
cin >> val1 >> val2 >> val3;
cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
return 0;
}

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Compile Result

```
Please enter your 3 numbers: 10 20 30
10
20
30
30
20
10

[Process completed - press Enter]
```

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TASK 04:

Formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

```
# include <iostream>
# include <cmath>
Using namespace std;
int main ( )
{
double x1, y1, x2, y2, distance;
Cout << "Enter x1 and y1:";
Cin >> x1 >> y1;
Cout << "Enter x2 and y2:";
Cin >> x2 >> y2;
distance = sqrt (pow (x2 - x1, 2) + pow (y2 - y1, 2));
Cout << "Distance = " << distance << endl;
   return 0;
}
```

Output:

Enter x1 and y1: 2 3

Enter x2 and y2: 5 7

Distance = 5

TASK 05:

Formula: Area of Rectangle = length × width

Area = Length × Width

```
# include <iostream>
Using namespace std;
int main ( )
{
 double length, width, area;
Cout << Enter length:";
Cin >> length;
Cout << Enter width:";
Cin >> length
       width;
area = length * width
Cout << Area of Rec
= " << area << endl
  return 0;
}
```

Area of Circle:

Formula: $A = PI \times r^2$

```
# include <iostream>
using namespace std;
int main ( )
{
double r, area;
Const double PI = 3.14159;
cout << "Enter radius: ";
cin >> r;
area = PI * r * r;
cout << Area of circle = " << area << endl;
   return 0;
   }
```

Area of Triangle: Area = Base x height x 0.5

```
# include <iostream>
using namespace std;
int main ( )
{
double base, height, area;
cout << Enter base:";
cin >> base;
cout << Enter height: ";
cin>> height;
area = base* height * 0.5;
cout << Area of Triangle = " area << endl;
   return 0;
}
```

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Coding C++

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RUN MENU

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    double x1, y1, x2, y2;
    double distance;

    cout << "Enter x1: ";
    cin >> x1;

    cout << "Enter y1: ";
    cin >> y1;

    cout << "Enter x2: ";
    cin >> x2;

    cout << "Enter y2: ";
    cin >> y2;

    distance = sqrt((x2 - x1) * (x2 - x1) +
                    (y2 - y1) * (y2 - y1));

    cout << "Distance = " << distance << endl;

    return 0;
}
```

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Compile Result

```
Enter x1: 2 3 5 7
Enter y1: Enter x2: Enter y2: Distance = 5

[Process completed - press Enter]
```

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Coding C++

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RUN MENU

```
    circleArea = 3.14159 * radius * radius;

    // Rectangle
    cout << "Enter length of rectangle: ";
    cin >> length;

    cout << "Enter width of rectangle: ";
    cin >> width;

    rectangleArea = length * width;

    // Triangle
    cout << "Enter base of triangle: ";
    cin >> base;

    cout << "Enter height of triangle: ";
    cin >> height;

    triangleArea = 0.5 * base * height;

    // Results
    cout << "\nArea of Circle = " << circleArea << endl;
    cout << "Area of Rectangle = " << rectangleArea << endl;
    cout << "Area of Triangle = " << triangleArea << endl;

    return 0;
}
```

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Compile Result

```
Enter radius of circle: 5 10 4 8 6
Enter length of rectangle: Enter width of r
ectangle: Enter base of triangle: Enter hei
ght of triangle:
Area of Circle = 78.5397
Area of Rectangle = 40
Area of Triangle = 24

[Process completed - press Enter]
```

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TASK 06:

Total purchase

Prices:

Item 1 = $ 12.95
Item 2 = $ 24.95
Item 3 = $ 6.95
Item 4 = $ 14.95
Item 5 = $ 3.95
Sales tax = 6%.

Output:

Item 1: $ 12.95
Item 2: $ 24.95
Item 3: $ 6.95
Item 4: $ 14.95
Item 5: $ 3.95
subtotal = $ 63.75
Sales tax = $ 3.83
Total: $ 67.58

```
#include <iostream>
#include <iomanip>
using namespace std;
int main()
{
double item1 = 12.95;
double item2 = 24.95;
double item3 = 6.95;
double item4 = 14.95;
double item5 = 3.95;
double subtotal;
double salestax;
double total;
subtotal = item1 + item2 + item3 + item4 + item5
salestax = subtotal * 0.06;
total = subtotal + salestax;
cout << fixed << setprecision(2);
cout << "Item 1: $" << item1 << endl;
cout << "Item 2: $" << item2 << endl;
cout << "Item 3: $" << item3 << endl;
cout << "Item 4: $" << item4 << endl;
cout << "Item 5: $" << item5 << endl;
cout << "subtotal: $" << subtotal << endl;
cout << "Sales tax: $" << salestax << endl;
cout << "Total: $" << Total << endl;
return 0;
}
```

Coding C++

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RUN MENU

```
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;

    double subtotal;
    double salesTax;
    double total;

    subtotal = item1 + item2 + item3 + item4 + item5;
    salesTax = subtotal * 0.06;
    total = subtotal + salesTax;

    cout << fixed << setprecision(2);

    cout << "Item 1 Price = $" << item1 << endl;
    cout << "Item 2 Price = $" << item2 << endl;
    cout << "Item 3 Price = $" << item3 << endl;
    cout << "Item 4 Price = $" << item4 << endl;
    cout << "Item 5 Price = $" << item5 << endl;

    cout << "Subtotal = $" << subtotal << endl;
    cout << "Sales Tax = $" << salesTax << endl;
    cout << "Total = $" << total << endl;

    return 0;
}
```

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Compile Result

```
Item 1 Price = $12.95
Item 2 Price = $24.95
Item 3 Price = $6.95
Item 4 Price = $14.95
Item 5 Price = $3.95
Subtotal = $63.75
Sales Tax = $3.82
Total = $67.58

[Process completed - press Enter]
```

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BONUS TASK:

Question: Input a five-digit integer and print individual digits with three spaces between them.

```
#include <iostream>
using namespace std;
int main ( )
{
  int number;
cout<< Enter a five-digit number: ";
  cin >> number;
cout<< number / 10000 << "   ";
cout<< (number / 1000) % 10 << "   ";
cout<< (number / 100) % 10 << "   ";
cout<< (number / 10) % << "   ";
 Cout << number % 10 << endl;
    return 0;
}
```

Output:

```
Enter a five-digit number:
12345
1   2   3   4   5
```

Home TASK:

1: Exchange / swap

Question: Read two variables and swap their values.

```
#include <iostream>
using namespace std;
int main ( )
{
 int var1, var2, temp;
cout<< Enter value of var1: ";
 cin >> var1;
cout<< Enter value of var2 : ";
 cin >> var2;
 temp = var1;
 var1 = var2;
 var2 = temp;
cout<< After swapping: " << endl;
cout << "var 1 = " << var1 endl;
cout << "var 2 = " << var2 endl;
cout << "var 3 = " << var3 endl;
 return 0;
}
```

Output:

```
Enter value of var1 = 10
Enter value of var2 : 20
After swapping
var 1 = 20
var2 = 10
```

Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int var1, var2, temp;

    cout << "Enter value of var1: ";
    cin >> var1;

    cout << "Enter value of var2: ";
    cin >> var2;

    cout << "\nBefore swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "\nAfter swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    return 0;
}
```

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Compile Result

```
Enter value of var1: 10 20
Enter value of var2:
Before swapping:
var1 = 10
var2 = 20

After swapping:
var1 = 20
var2 = 10

[Process completed - press Enter]
```

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Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int number;

    cout << "Enter a five-digit number: ";
    cin >> number;

    int digit1 = number / 10000;
    int digit2 = (number / 1000) % 10;
    int digit3 = (number / 100) % 10;
    int digit4 = (number / 10) % 10;
    int digit5 = number % 10;

    cout << digit1 << "    "
         << digit2 << "    "
         << digit3 << "    "
         << digit4 << "    "
         << digit5 << endl;

    return 0;
}
```

CamScanner

Compile Result

```
Enter a five-digit number: 12345
1   2   3   4   5

[Process completed - press Enter]
```

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Home Task 2: Make Upper

Question:

```
#include <iostream>
using namespace std;
int main ( )
{
char ch;
cout << Enter a lower case alphabet: ";
cin >> ch;
ch = ch - 32;
cout << uppercase alphabet :" << ch << endl;
return 0;
}
```

Output:

Enter a lowercase alphabet : j
uppercase alphabet : J

Coding C++

Auto saved at 21:47:59

RUN MENU

```cpp
#include<iostream>
using namespace std;

int main()
{
  char letter;
  cout<<"Enter a lowercase alphabet: ";
  cin>>letter;
  letter=letter - 32;
  cout<<"Uppercase alphabet:"
  <<letter<<endl;
  return 0;
  }
```

CamScanner

Compile Result

```
Enter a lowercase alphabet: b
Uppercase alphabet:B

[Process completed - press Enter]
```

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